

PRD

BPM C2 - Group R | Assignment 5

Gmail Language Translation

Team: Gmail Team

Contributors: SPMs, Engineer, Director

Resources: Designs, Research, Notes

Status: Solution Review

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Objective: Improve Google's retention of non-English speaking users (both in English speaking and non-English speaking countries) by enabling them to translate message content in Gmail to other languages of their choice, thereby making the email platform easier to use for non-English speaking users.

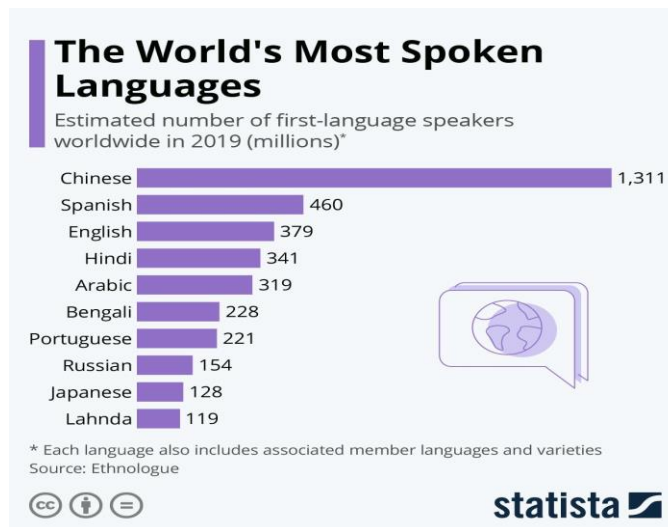
Problem Alignment

Gmail is only supported in the English language as of today. Of the world's 195 countries, close to 86 countries have English as their official language or are widely spoken. But it also means that 56% of the countries speak a non-English language or do not have proficiency in English as a medium of communication. ([Link](#))

And in some cases, even though English might be an official language, not every citizen might be proficient in English or be comfortable with it as a conversational language. For e.g. in India, close to 122 languages are spoken. ([Link](#)). There are many countries especially in South Asia where the local language is the primary language of business - for example Japan, South Korea and China wherein users converse in their local languages for business and communication. As Gmail is not offered in non-english languages, they are potentially losing out on the non-English population altogether.

Why are we building this - core pain points

- a. A Gmail user who does want to view his emails and messages in a non-English language is unable to do so. The user has to use other tools such as Google Translate to view the message in their local language
- b. A Gmail user cannot compose and send an email in a non-English language and has to again rely on other tools such as Google Translate to do so, thereby making the email process inconvenient
- c. Since email is a two-way communication between a sender and a receiver, if the translation in non-English language is a pain point then it affects the user experience on both sides. Hence it is important to solve the pain point on both sides - the sender and receiver.



There are several use cases why users translate while their use of a product or service:

- A. Getting around in the app/ web page - to know more about the services/ availability aka casual browsing (Individual users)
- B. Translating important information like - Travel maps during car rides, tax documents, user manual, medical document (Enterprise usage)
- C. Voice recognition software - Real time decoding for e.g. during student lectures, client webinars in MNCs, culinary recipes or real estate tours or hospitality sectors which entertains multi nationalities, or for World Cup fans who attend the event live, where you need live translations (Enterprise usage)

1. Goals of Translation in Gmail

- 1.1. Increase usage of Gmail among non-English speaking population
- 1.2. Improve retention of Gmail users
- 1.3. Increase market share in enterprise email usage in non-English speaking countries/ population

2. **Competition** - The main competitor we feel in this space is Microsoft. Below Mind Map shows how they are competing with Google in this space. Microsoft's email client Outlook, supports translation in other languages. Outlook is primarily used in the Enterprise space though.



Do we use Google Translate or using some other technology -Yes we are embedding Google Translate into Gmail to translate the desired content as interoperability with Gmail would be easier given that both are Google products.

Analysis of Google Translation accuracy:

- When text is well-structured, written using formal language, with simple sentences, relating to formal topics for which training data is ample, it often produces conversions similar to human translations between English and a number of high-resource languages.
- Human evaluation from English to other languages shows that the main idea of a text is conveyed more than 50% of the time for 35 languages. ([Reference Link](#))

Key features

Key features of the proposed solution include the following:

- i. Email messages on Gmail to be translated into users' local language when it arrives in the inbox (mockup below)
- ii. Email messages could be composed in the local language (preference) of the user (mockup below)
- iii. The settings and support for Gmail should be in the local language of the user (mockup below)
- iv. The user has the ability to change the local language preference on Gmail
- v. The user's existing messages in the inbox are translated into the local language preference of the user.
- vi. The different menu options that appear on Gmail - such as Inbox, spam, drafts etc should all be translated in the local language
- vii. Users should be able to search emails in both the local language and translated language
- viii. The translation of email should work on both the mobile apps and desktop for Gmail. The Gmail app should translate messages in the local language on both iPhone and Android devices

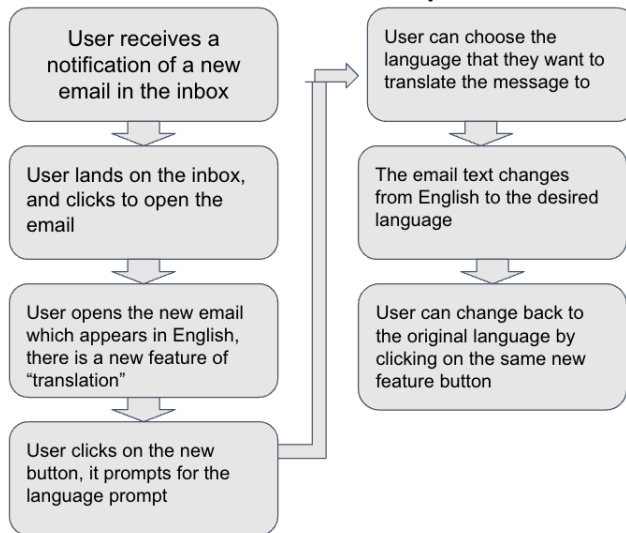
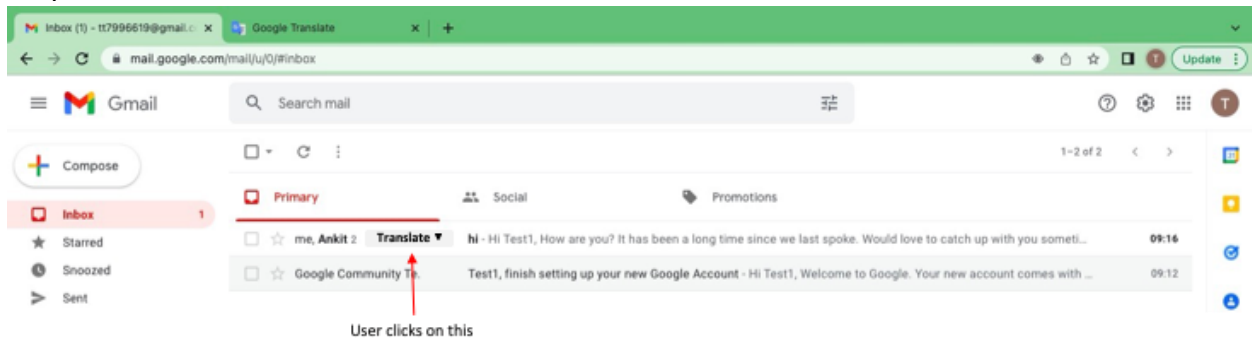
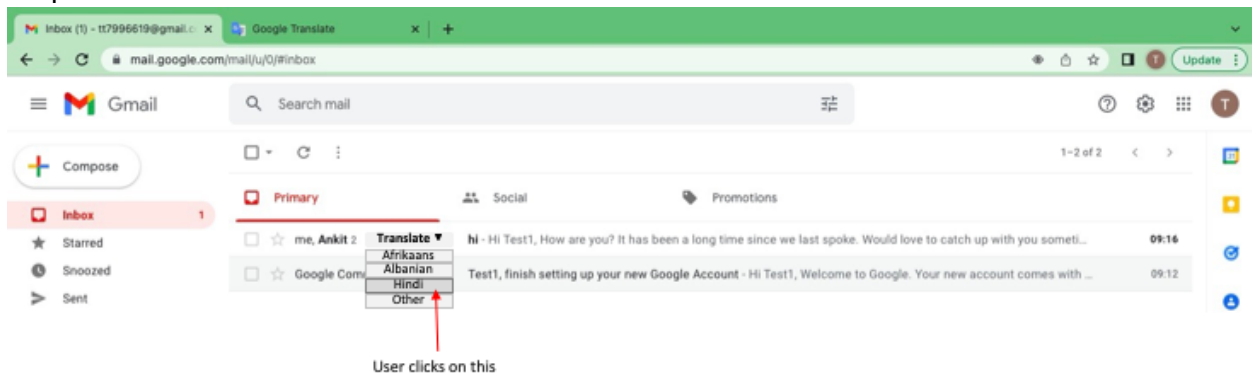
Items in future scope

1. The apps within the Gmail ecosystem are translated into the local language for the user
2. The translation feature should work both for Gmail for end-user and Enterprise usage

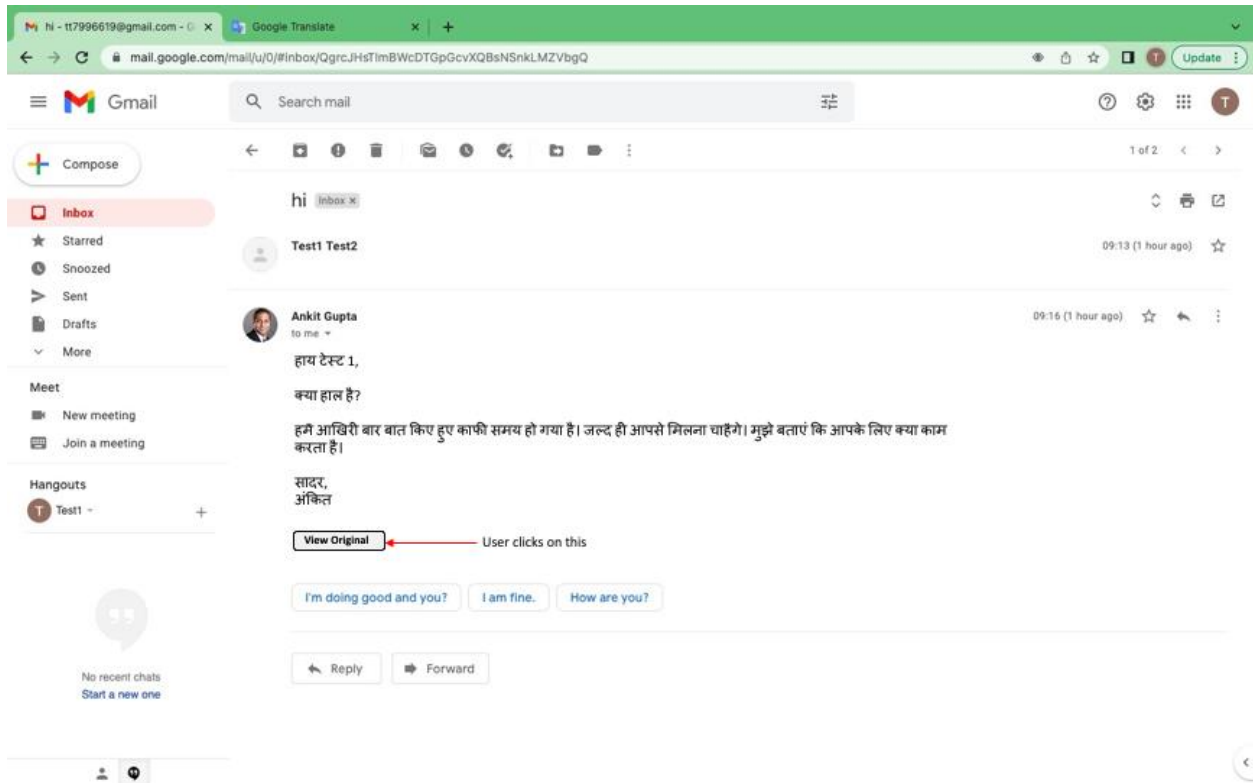
Mockups of Key Features:

- i. Email messages on Gmail to be translated into users' local language when it arrives in the inbox

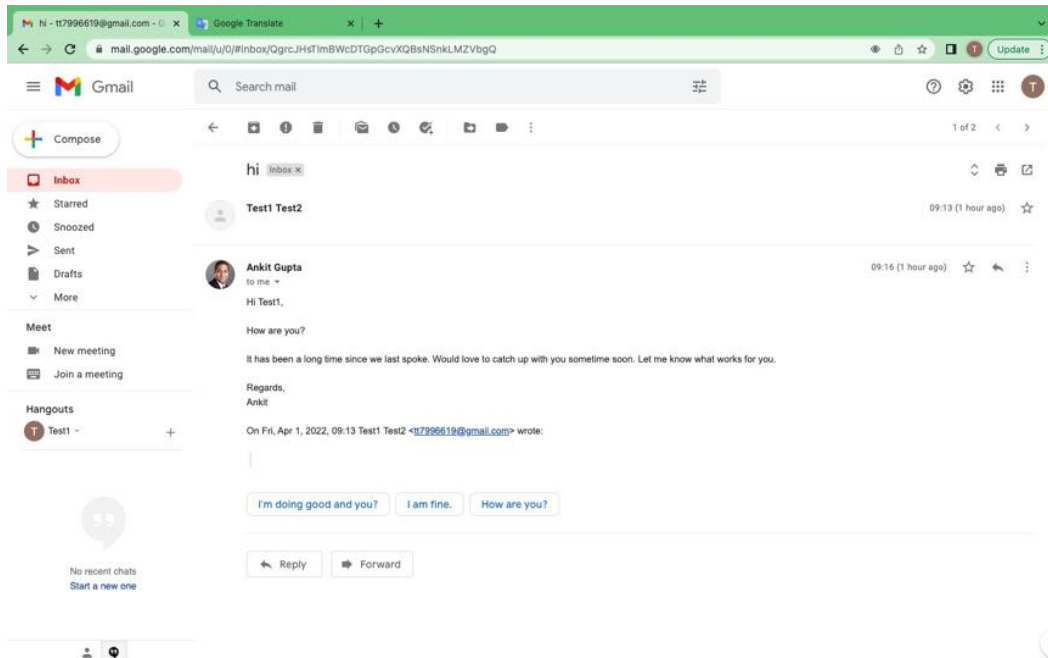
Workflow of new email arrival, after the release of the translation feature (First time use)

**Step 1:****Step 2:**

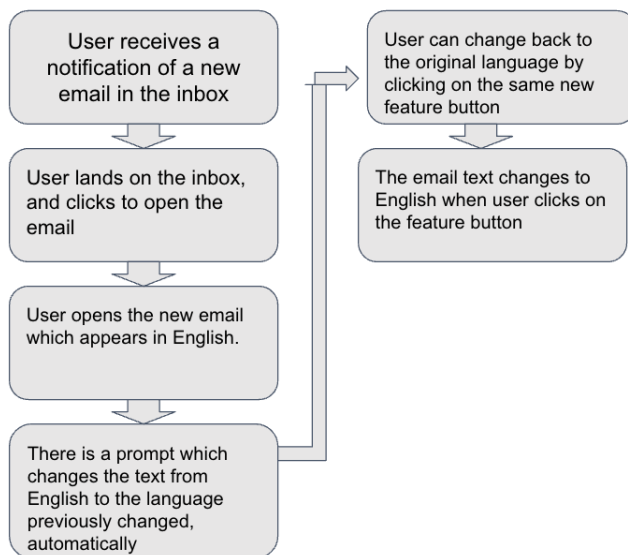
Step 3:



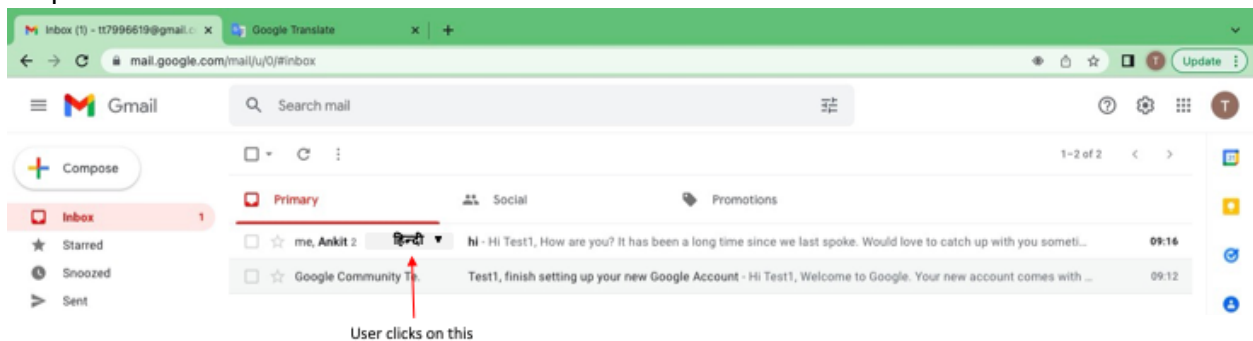
Step 4:



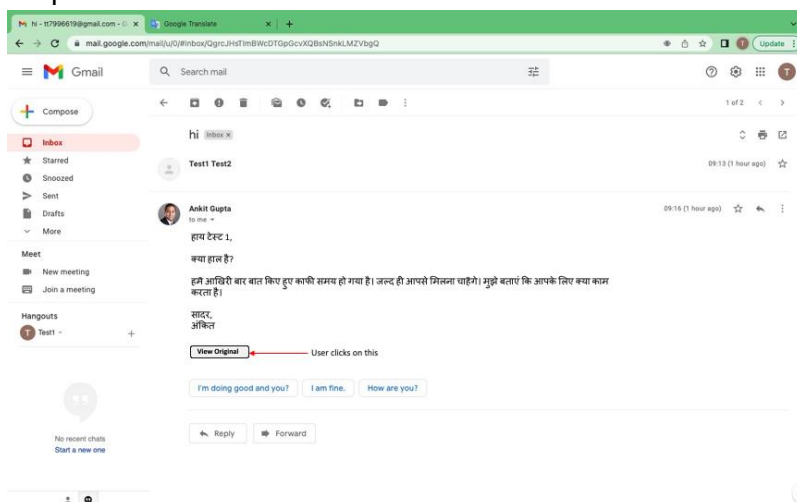
Workflow of new email arrival - repeat usage



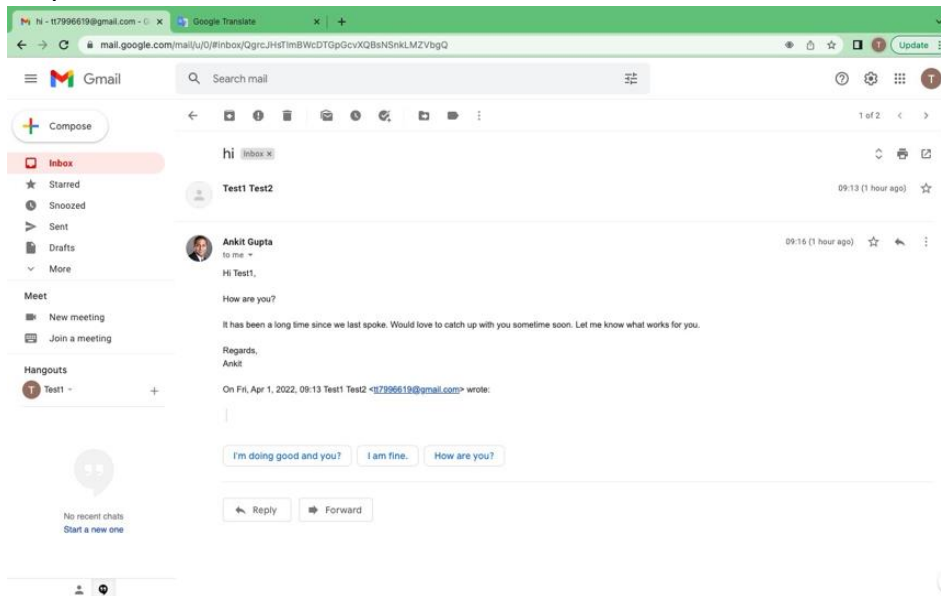
Step 1:



Step 2:

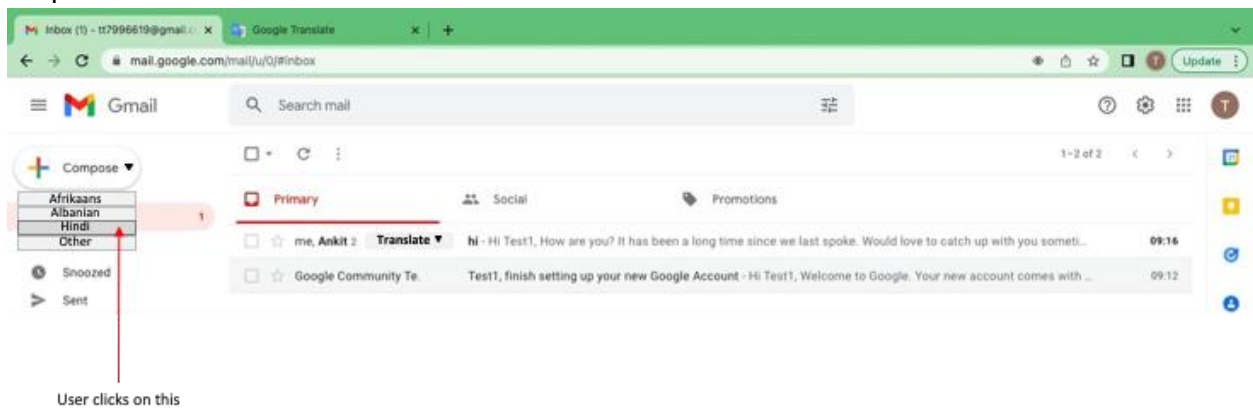


Step 3:

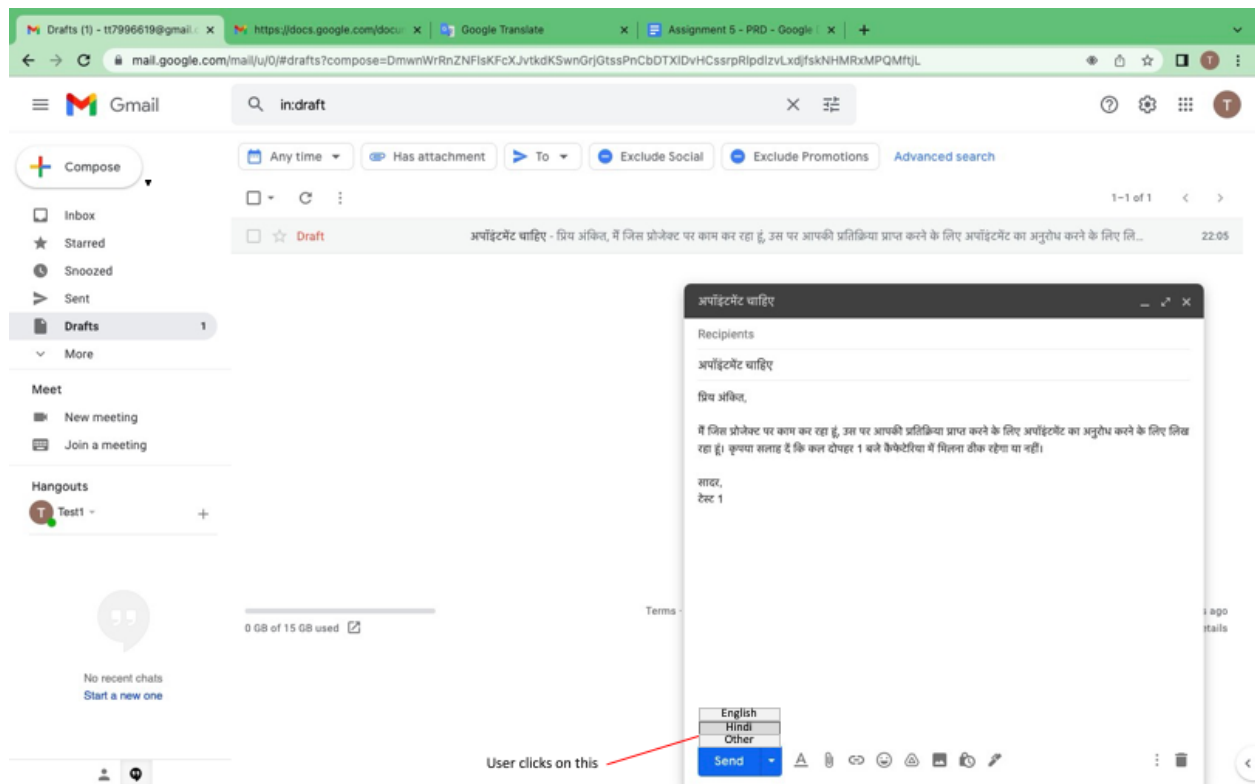


ii. Email messages could be composed in the local language (preference) of the user

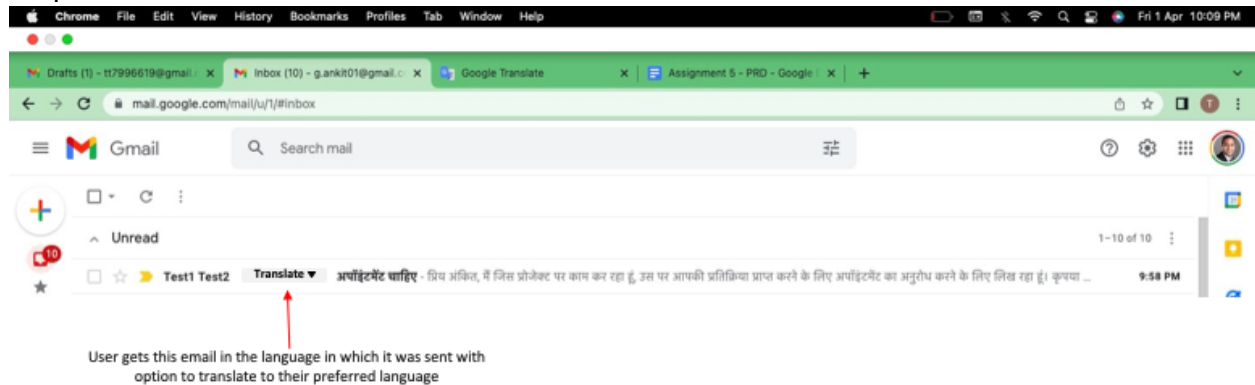
Step 1:



Step 2:



Step 3:



iii. The settings and support for Gmail should be in the local language of the user



Prioritization and Metrics

User features	Priority	Metrics
1. Email messages could be composed in the local language (preference) of the user	P0	# of users who used the translation feature in Gmail to a different Non-English language
2. The user has the ability to change the local language preference on Gmail	P0	# of users who changed their preference to non-English language # of times user changed preference back to English from another language (to measure how useful is the feature for the user)
3. The user's existing messages in the inbox are translated into the local language preference of the user.	P1	# of users who continue to actively use Gmail (7D period) with the translated language # of users who changed their preference to non-English language
4. The different menu options that appear on Gmail - such as Inbox, spam, drafts etc should all be translated in the local language	P3	# of users who continue to actively use Gmail (7D period) with the translated language
5. Users should be able to search emails in both the	P2	# of users and # of queries searched on Google in the

local language and translated language		non-English language # of users who used search multiple times in non-English language and then changed search to English (to measure accuracy of the search)
6. The settings and support for Gmail should be in the local language of the user	P4	# of users who continue to actively use Gmail (7D period) with the translated language
7. The translation of email should work on both the mobile apps and desktop for Gmail. The Gmail app should translate messages in the local language on both iPhone and Android devices	P0	# of users who changed language preference to non-English on Mobile device - both Android and iOS (7D period) # Gmail had latency issues on mobile app due to language in non-English

Among the 1, 2, 7, 1st Feature will be given a priority over others because it's the basic easy to implement and important feature.

Privacy concerns - Gmail will not sample or store the data of the user.

FAQs or questions

This section is for the audience - developers/ QA/ sales/ marketing etc. who will be reading this document.

Person	Question

Study Presented By-

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